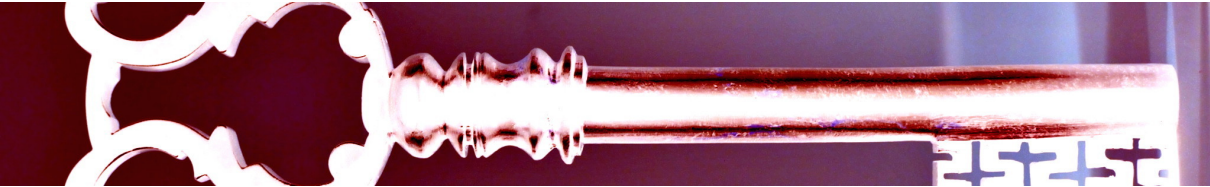
universität innsbruck



Cryptography

Rainer Böhme

5 March 2026

Agenda

1. **Course organization**
2. Basic concepts
3. Cryptographic keys

Organization

This class belongs to the 3-out-of-6 elective modules in the MSc Computer Science.

This module in the Summer Term 2026					
Lecture	VO	Thu 08:30–10:00	HSB 2	Rainer Böhme	4.5 ECTS
Proseminar	PS	Tue 08:30–10:00	HSB 9	& Verena Lachner	3.0 ECTS
					7.5 ECTS

- Attendance is voluntary for the lecture and mandatory for the proseminar.
- The lecture is graded in a written exam (25 June); second attempt: mid-September
- We encourage active participation. There will be no recordings of in-person classes.

Learning Goals

- Students master the concepts and theoretical basics for the construction of current cryptographic procedures.
- They know elementary security terms and typical attacks on cryptography.
- They can analyse the secure use of selected cryptographic procedures in practical systems and, if necessary,
- familiarise themselves with similar procedures independently.

What this course is not

- "Crypto" means cryptography, not cryptocurrency.
- The course does not teach you how to securely implement cryptography.
- It is by far not comprehensive enough to let you design your own cryptographic methods. We **strongly advise against**, and the course should teach you why.
- It provides the foundations for information security and privacy techniques, but it is not a course on security & privacy. → Join VO 703503 in the winter term.

Cryptographic Primitives

Coverage across selected modules we teach in Innsbruck

	Applied Cryptography BSc Computer Science	Software Security Engineering MSc Software Engineering	Cryptography MSc Computer Science
Cryptographic hash	✓	✓	✓
Symmetric block cipher	✓	✓	✓ ✓
Asymmetric encryption	✓	✓	✓ ✓
Digital signature	✓	✓	✓ ✓
Commitment scheme			✓ ✓
Others	✓		✓

Legend: ✓ = abstract interface; ✓ = security definition; ✓ = concrete algorithms

Scientific Approach to Cryptography

Questions we ask:

- How can we talk about security, a non-functional property, in a precise way?
- How can we meaningfully define security properties?
- How can we construct systems, in a principled way, that meet our definitions?
- How **confident** can we be even when our systems are under **stress**?

Subjects we study:

- (Meanwhile) academic examples: one-time pad, information-theoretic MACs, DES
- Real-world examples: DH, RSA, AES, ECDH, ECDSA
- A few black boxes: SHA2, Keccak, . . .

Mindset: Anticipation of attacks. Pessimism. Professional paranoia.

Prerequisites

How to put randomness into equations?

→ Probability theory

What is Shannon information?

→ Information theory

What is the difference between source and channel coding?

→ Coding theory

What is the purpose of the extended Euclidean algorithm?

→ Number theory

What does “NP complete” mean?

→ Complexity theory

Syllabus

Summer Term 2026

05.03.26	1.	Introduction, basic terminology
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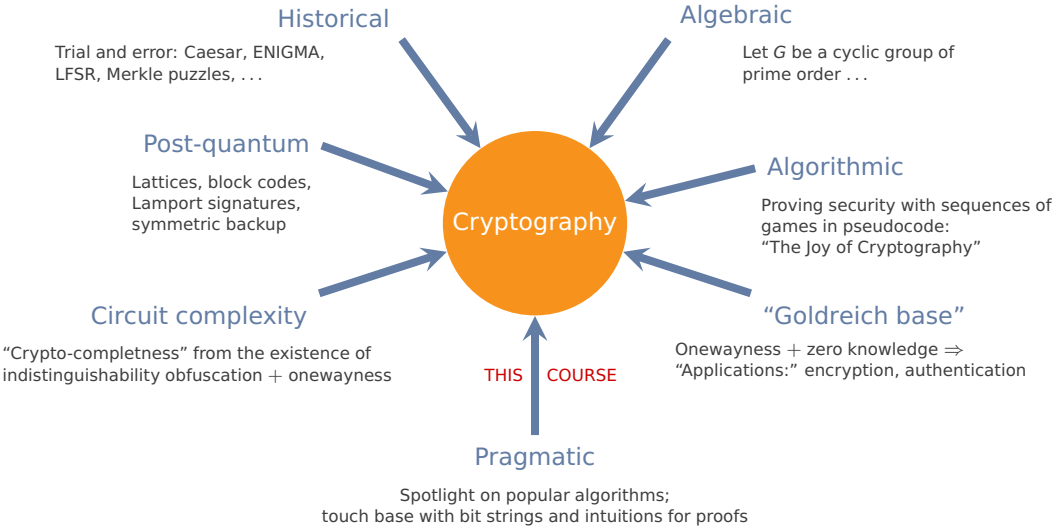
25.06.26		**Written exam**

Roads to Cryptography



Resources

Recommended textbooks

- J. Katz & Y. Lindell (2021): Introduction to Modern Cryptography, 3rd ed., CRC Press
- C. Paar & J. Pelzl (2010): Understanding Cryptography, Springer
- M. Rosulek (2025): The Joy of Cryptography, MIT Press
- S. Vaudenay (2005): A Classical Introduction to Cryptography, Springer

Online resources

- The English Wikipedia is often a good starting point.

  Hence, LLMs are good at concepts, but they often get numerical and symbolic examples wrong.
- We are maintaining a public resource page for this course with references to original literature.

  http://informationsecurity.uibk.ac.at/teaching/infosec/
- Also read the general FAQ under Teaching.

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Communications Security

e. g. systematic eavesdropping

We distinguish:

- Protection of communication contents
- Protection of communication circumstances (traffic data)

Beware of Security Jargon

*"The term **'attack'** is used here in a technical sense that differs somewhat from common English usage.*

In common English usage, an attack is an aggressive action perpetrated by an opponent, intended to enforce the opponent's will on the attacked party.

The term is used here to refer to behavior that subverts the intent of communicating parties without the agreement of those parties."

RFC 7258 (2014)

Basic Protection Goals

Confidentiality	Information is accessible to authorized parties only.
Integrity	Modification of information can be detected.
Availability	Authorized parties can access information or use resources when and where needed.

Information	can be system state or architecture (software and hardware)
Party	can be person(s) or computers under their control

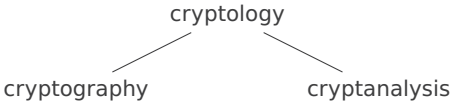
Enter Cryptography

Mechanisms to enforce the basic protection goals in communication systems:

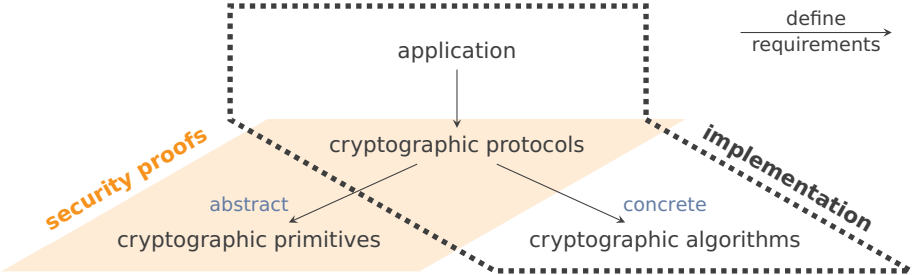
Protection goal	Mechanism	
Confidentiality	Encryption	Cryptography
Integrity	Authentication	
Availability	Redundancy + Diversity	

Cryptography and the Real World

Ontology



Composition



Hilbert's Program for Cryptography ?

How a standard textbook characterizes modern cryptography:

- **Formal definitions of security** are an essential first step in the design of any cryptographic primitive or protocol.
- **Precise assumptions:** security often relies on some widely believed, but unproven assumption(s).

  Any such assumption must be clearly stated and unambiguously defined. By relying on long-standing assumptions, it is possible to design schemes that are unlikely to be broken.
- **Rigorous proofs of security** guarantee that schemes are secure unless the underlying assumptions are false or the security definition did not appropriately model the real-world security concerns.

Katz & Lindell 2014, p. xv, emphasis added

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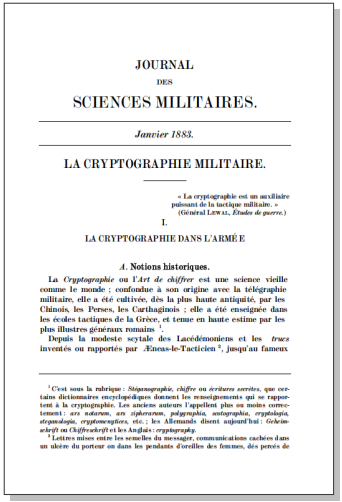
Kerckhoffs' Principle

Cryptographic algorithms must be public.

The security depends on the secrecy of keys **only**.

Rationales

- Finding [changing] secure keys is easier than finding [changing] a secure system
- Err on the side of caution
- Avoid blind trust
- More eyes find more flaws
- The only reasonable assumption when protecting public infrastructures



JOURNAL

DES

SCIENCES MILITAIRES.

Janvier 1883.

LA CRYPTOGRAPHIE MILITAIRE.

« La cryptographie est un auxiliaire puissant de la tactique militaire. »
(Général LEWAL, *Études de guerre.*)

I.

LA CRYPTOGRAPHIE DANS L'ARMÉE

A. **Notions historiques.**

La *Cryptographie* ou l'*Art de chiffrer* est une science vieille comme le monde ; confondue à son origine avec la télégraphie militaire, elle a été cultivée, dès la plus haute antiquité, par les Chinois, les Perses, les Carthaginois ; elle a été enseignée dans les écoles tactiques de la Grèce, et tenue en haute estime par les plus illustres généraux romains [1].

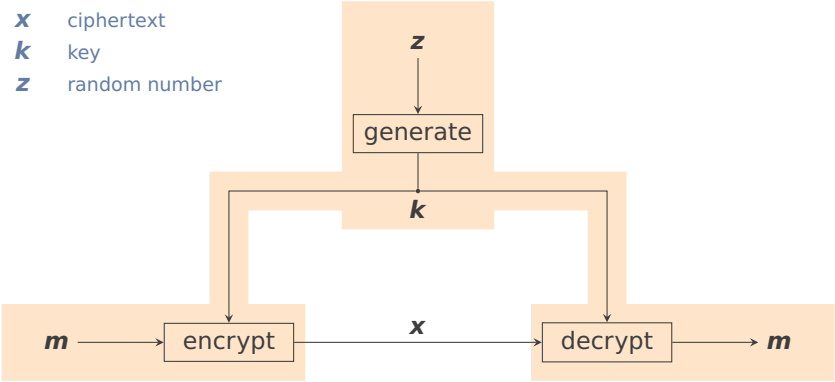
Depuis la modeste scytale des Lacédémoniens et les *trucs* inventés ou rapportés par Æneas-le-Tacticien [2], jusqu'au fameux

[1] C'est sous la rubrique : *Stéganographie, chiffre* ou *écritures secrètes*, que certains dictionnaires encyclopédiques donnent les renseignements qui se rapportent à la cryptographie. Les anciens auteurs l'appellent plus ou moins correctement : *ars notarum, ars ciphrarum, polygraphia, autographia, cryptologia, steganologia, cryptomenytica*, etc. ; les Allemands disent aujourd'hui : *Geheimschrift* ou *Chiffrescrift* et les Anglais : *cryptography*.

[2] Lettres mises entre les semelles du messager, communications cachées dans un ulcère du porteur ou dans les pendants d'oreilles des femmes, dés percés de

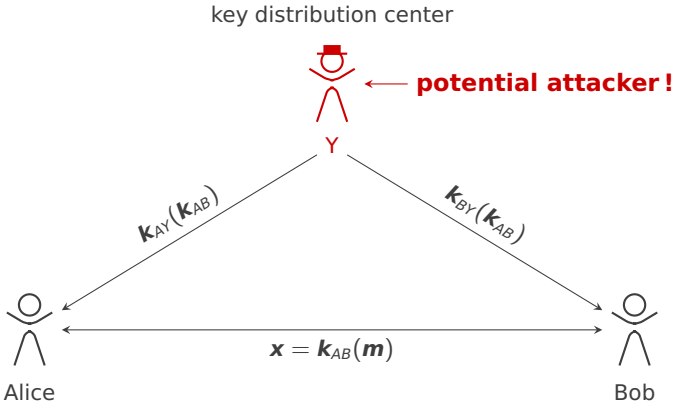
Symmetric Encryption

m message
x ciphertext
k key
z random number



trusted area

Distribution of Symmetric Keys



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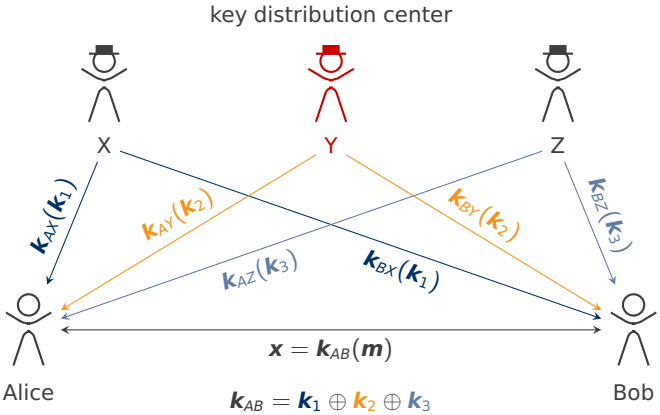
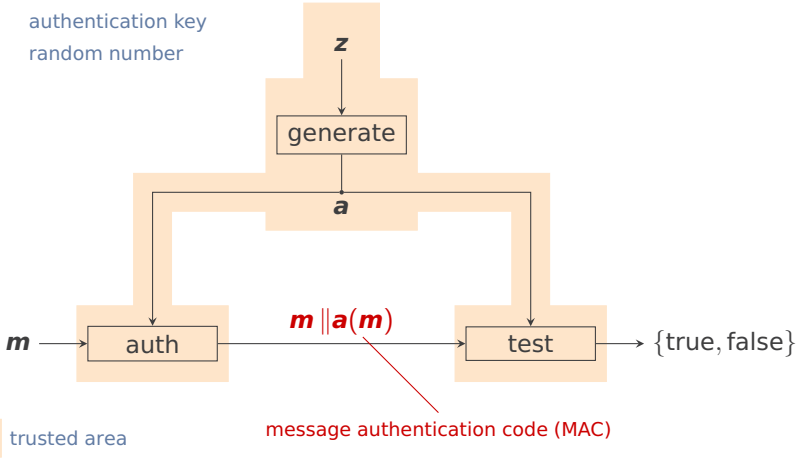


→ **Distributed trust** as security principle: no single powerful party

Symmetric Authentication

$\boldsymbol{m}$ message
$\boldsymbol{a}$ authentication key
$\boldsymbol{z}$ random number



Role of Cryptographic Keys

- Knowledge of keys is the only feature that tells defenders and attackers apart.
- In this sense, keys make authorization decisions for encrypted bits “on the wire” (which, unlike systems with I/O, cannot be programmed to make authorization decisions).
- The objective of many cryptographic algorithms is to leverage the defenders’ small information advantage (a few key bits) to protect large secrets (gigabytes of data).
- Key generation, exchange, and association to parties are crucial for security.

→ Cryptography is the science of secrets, that is the [proven] **absence** of information.

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